

Kamal Swarnkar

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SKILLS

Languages:	Python, C
Tools:	Git, Linux, Docker, Postman, VSCode, Celery, JWT, SolidWorks, OpenRocket
Libraries & ML:	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, spaCy, NLTK
Frameworks:	Django, Django REST Framework
Databases:	PostgreSQL, Redis
Deployment:	Vercel, Render, Railway
Soft Skills:	Self-motivated, Articulate, Organized, Leadership

PROJECTS

AI Career Lab | GitHub | Live Demo

Mar' 2026

- Architected an ML-based career guidance platform using **TF-IDF** and **multi-class classification**, achieving **86.58% accuracy** with minimal overfitting (1.23% gap).
- Evaluated Random Forest, SVM, and Logistic Regression using **precision, recall, and F1-score**; best model achieved **86.56% F1-score on unseen data**.
- Designed a robust training pipeline with explicit **train-test gap analysis** to ensure strong **generalization** and prevent overfitting.
- Integrated ML inference pipeline with **joblib**-based model loading and **real-time prediction** for career recommendations.
- Designed a synchronous request-response workflow combining ML predictions with AI-generated insights, ensuring reliable **output validation** and **fallback handling**.
- Built a lightweight server-rendered backend using Django templates with optimized **static asset delivery** for efficient **deployment**.
- **Tech Stack:** Python, Django, Scikit-learn, Pandas, NumPy.

Resumix - Resume Analysis & Role Prediction Platform | GitHub | Live Demo

Feb' 2026

- Built a production-ready Django + DRF backend with modular **ViewSets** for authentication, resume analysis, history tracking, and **admin analytics**.
- Implemented stateless authentication using **JWT (SimpleJWT)**, enabling secure and scalable **sessionless API** access.
- Integrated asynchronous task processing using **Celery** with **Redis** as broker and result backend to offload AI-based suggestion generation.
- Improved system responsiveness by executing heavy ML enrichment tasks **asynchronously** while returning primary analysis **results in real time**.
- Engineered a weighted scoring pipeline (**TF-IDF**, skills, experience, ATS) with automated **gap detection** and recommendations.
- Optimized database queries using **indexing** and relation **prefetching** to reduce latency and improve performance under concurrent usage.
- **Tech Stack:** Python, Django, PostgreSQL, JWT, Scikit-learn, XGBoost, spaCy, NLTK, Celery, Redis.

TRAINING

FLAMES '25 MERN with Gen AI | W3grads | Certificate

Jul' 2025

- Addressed major development bottlenecks—skill gaps, AI inconsistencies, database lag, and deployment issues; **streamlined workflows** and optimized processes, leading to faster development, reliable AI performance, and **stable deployments**.
- Encountered collaboration hurdles like merge conflicts and time-zone delays; established a strict **Git workflow** and daily **SCRUMs**, which reduced conflicts and kept the team aligned.

Youth Research Program | Resolute Lab | Certificate

Mar' 2025

- Applied aerospace engineering concepts to design and simulate rocket systems using **OpenRocket** and **SolidWorks**; performed propulsion analysis, payload integration studies, and structural feasibility simulations.
- Collaborated in a multidisciplinary team to design a high-power rocket with simulated payload, optimizing parameters via **iterative simulations** and maintaining precise **technical documentation**.

CERTIFICATES

- Master Generative AI & Generative AI tools (Infosys) | Certificate Aug' 2025
- Build Generative AIApps and Solutions with No-Code Tools | Certificate Aug' 2025
- The Bits and Bytes of Computer Networking (Google) | Certificate Sep' 2024
- Computer Communications (University of Colorado) | Certificate Sep' 2024

ACHIEVEMENTS

- Qualified GATE 2026 (CS/IT) in first attempt with a score of 383 with AIR 21411
- Solved 100+ DSA Problems on LeetCode.
- MEDHA 2024 Hackathon: Achieved 10th rank among 60 teams.

EDUCATION

Lovely Professional University	Phagwara, Punjab
• Bachelor of Technology - Computer Science and Engineering – CGPA: 8.87	Aug' 2023 – Present
Central Academy Senior Secondary School	Bhilwara, Rajasthan
• Intermediate – 91%	Apr' 2021 – Mar'2022
• Matriculation – 89%	Apr' 2019 – Mar'2020